How Do Chunk Size, Chunk Overlap, and Retrieval Depth Affect Answer Quality and Computational Latency in a Locally Deployed Retrieval-Augmented Generation System on Consumer Hardware?

Prajjwal Singh
Independent Technical Researcher
Lucknow, India
prajjwals.acad@gmail.com

***Abstract*—This study evaluates the architectural performance trade-offs of localized, privacy-preserving Retrieval-Augmented Generation (RAG) system pipelines deployed on consumer-grade hardware sandboxes. Utilizing Python, LangChain, ChromaDB, and an offline Llama 3 instance via Ollama, we systematically measure the impact of text token segmentation boundaries on data indexing times and generation inference cycles. By executing 15 discrete experimental evaluation trials against a structured multi-category testing repository, we map the exact non-linear frontier separating resource latency from model grounding. Our results confirm the viability of zero-cost, localized document intelligence platforms while establishing parameter optimization boundaries.**

***Index Terms*—Retrieval-Augmented Generation (RAG), Large Language Models, Computational Latency, ChromaDB, Ollama, Performance Optimization.**

I. Introduction

Centralized, cloud-hosted Large Language Model (LLM) architectures expose sensitive materials to network dependencies, substantial token subscription fees, and data privacy vulnerabilities. While Retrieval-Augmented Generation (RAG) mitigates systemic model hallucinations by injecting localized database text segments directly into the model context window, contemporary deployments remain heavily tethered to cloud API services. This study addresses these vulnerabilities by engineering a 100

II. Experimental Setup & Methodology

The system architecture was split into modular components to evaluate performance across a structured dataset:

- **Ingestion Processing:** Document parsing via PyPDF, stripping out page layouts and embedding raw text segments into a local instance of ChromaDB utilizing a HuggingFace `all-MiniLM-L6-v2` matrix generator.
- **Grounding Constraints:** To enforce zero-hallucination behavior, the model prompt window was restricted to accept responses exclusively supported by retrieved context nodes, returning a standard fallback statement if data was absent.
- **Hardware Software Environment:** All testing loops were executed locally on an Apple laptop configured with an **Apple macbook air M5** processing architecture, running **16 GB** of unified RAM, macOS, Python 3.14, and local Llama 3 weights orchestrated via Ollama.

III. Empirical Evaluation & Performance Analysis

To capture accurate resource variations and clear out background system measurement noise, we executed an automated python benchmark script looping through 5 distinct text segmentation scales (200, 400, 600, 800, and 1000 tokens), running 3 distinct tracking trials per question block across a controlled 4-category test pool.

A. Ingestion Latency vs. Inference Synthesis

Our empirical observations show a distinct inverse relationship between document ingestion processing and real-time inference intervals. When configuring smaller token chunk blocks (200 tokens), the system generates an expanded number of matrix segments. This directly multiplies vector calculation density, which extends total database initialization times.

Conversely, larger structural chunk arrays (1000 tokens) lower structural indexing processing times significantly. However, they force the underlying local Llama 3 weights to process much larger context sizes during query execution, which can degrade answer quality if too much irrelevant text is fed into the prompt window.

IV. Limitations & Future Scope

The evaluations conducted during this study were bounded by a single open-source model lineage (Llama 3) running on isolated local consumer laptop hardware. Performance tracking vectors may change if deployed over hyper-dense corporate server layers or alternative asymmetric multi-model network clusters. Future expansions will explore dynamic, non-linear chunk-size shifting engines.

V. Conclusion

Our research confirms that fully localized, offline RAG pipelines provide highly secure, zero-cost academic document evaluation frameworks. This setup eliminates corporate data exposure risks while providing a stable balance between system resource consumption and text processing accuracy.